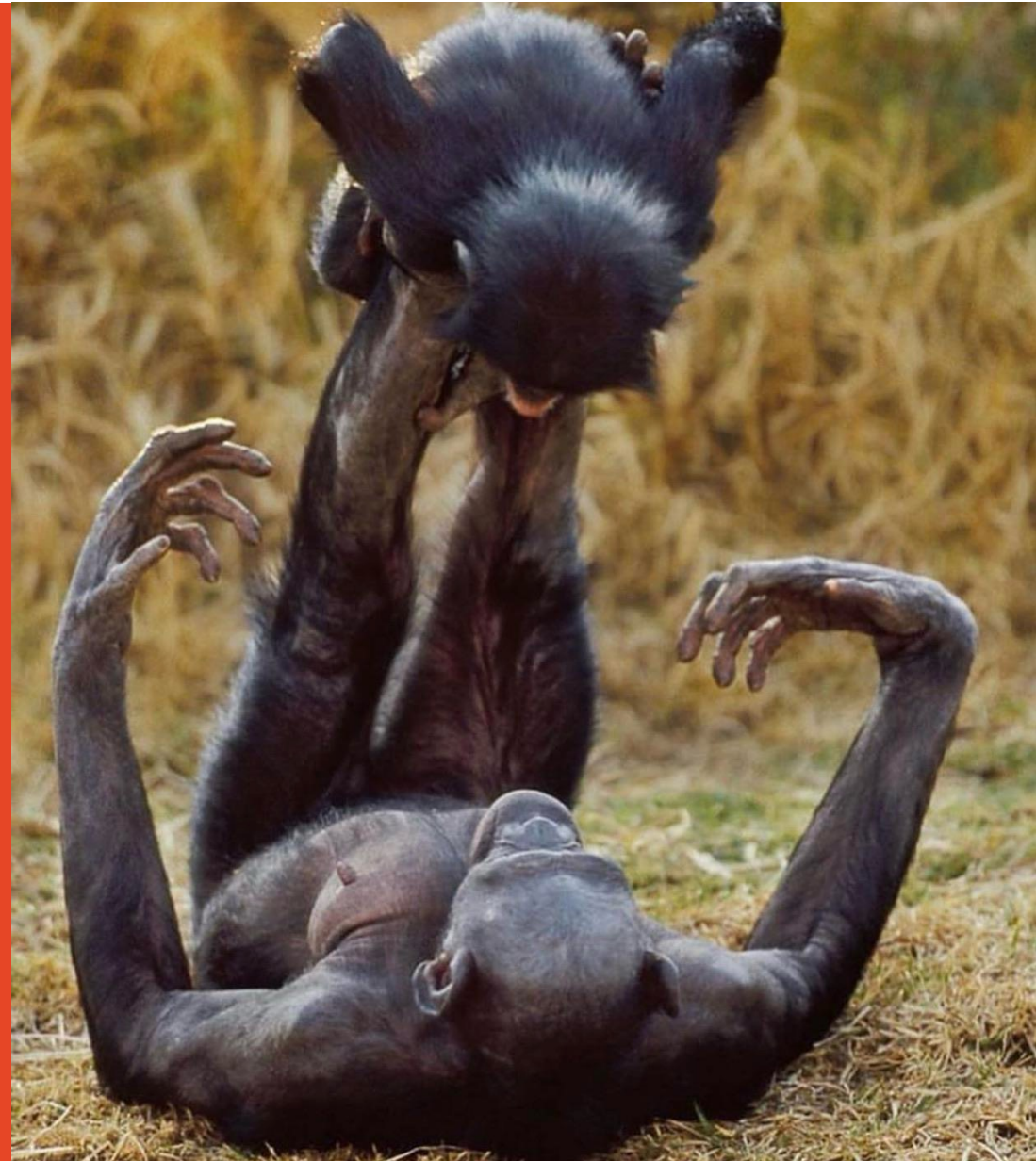


The Biological Basis of Behaviour

Ian Johnston
PSYC1003



Learning outcomes

Describe Darwin's theory of evolution by natural selection

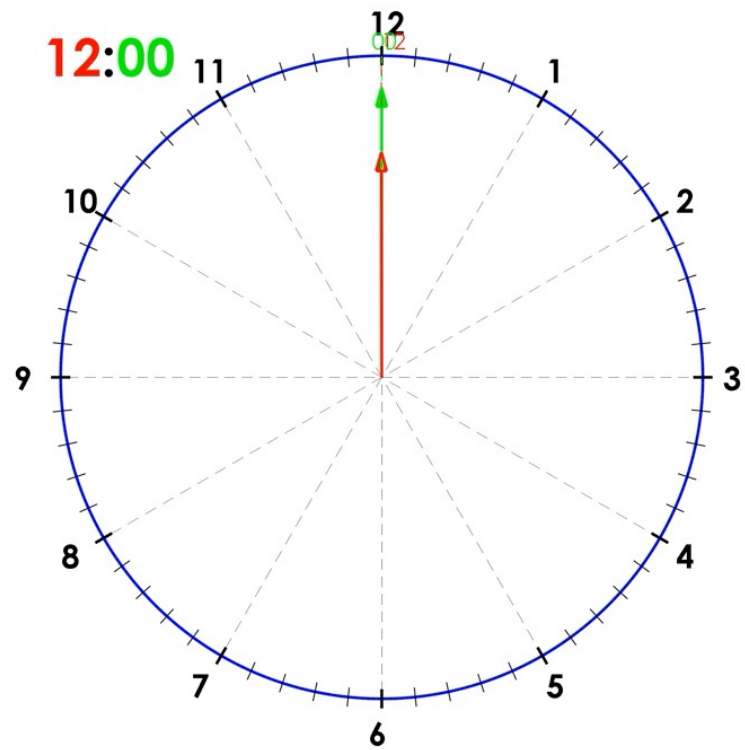
- Define fitness in evolutionary theory
- Define the three essential components of this theory

Evidence for the evolution of mental abilities:

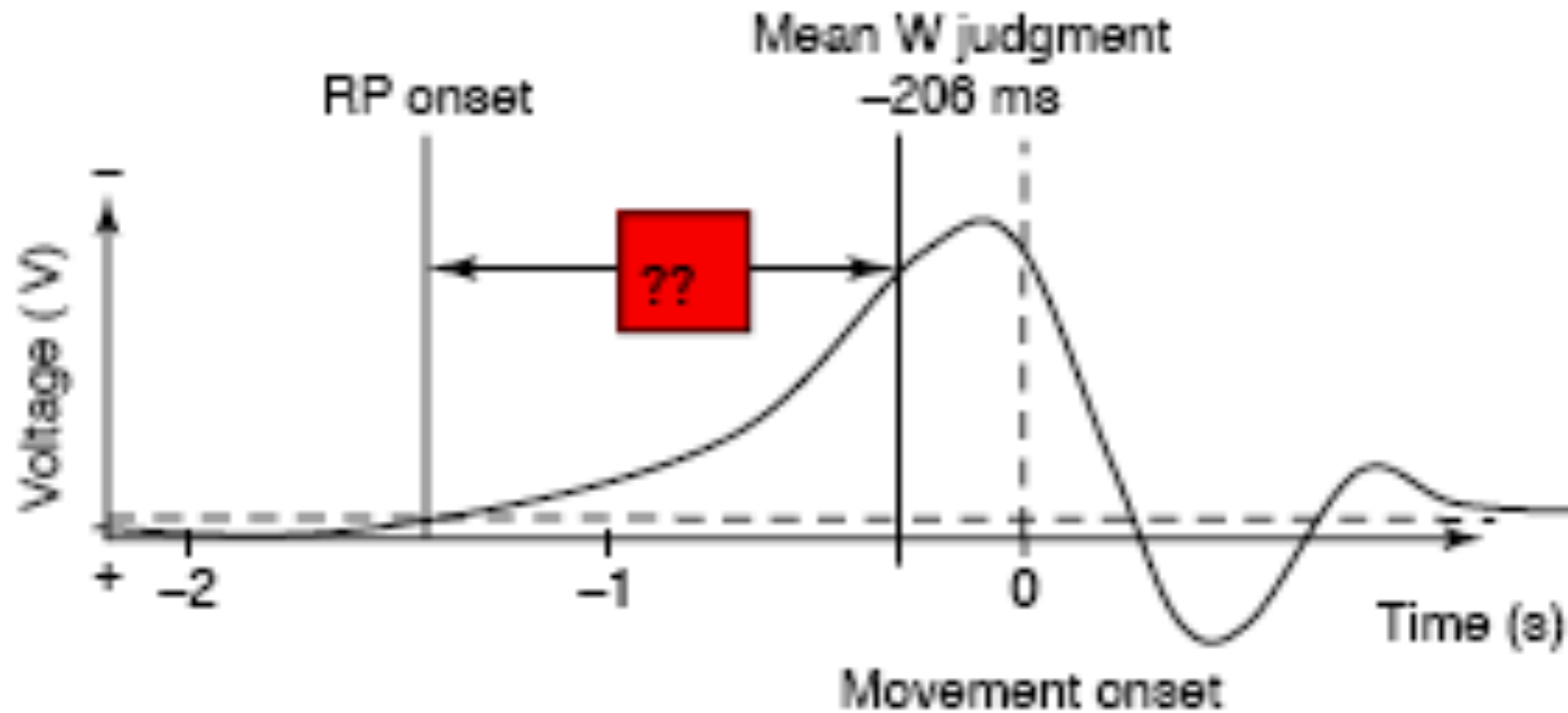
- Comparative psychology and neuroanatomy
- Behaviours affect fitness
- Artificial selection of mental abilities

Describe inclusive fitness

What causes our behaviour?



Is behaviour intentional?



Libet et al. (1983) *Brain*, 106: 623-642

Behaviour

Our behaviour is critical for satisfying our important biological needs:

4 Fs

Feeding (appetitive motivation, growth)

Fighting (aggression)

Fleeing (aversive motivation, safety, protection)

Sexual behaviour (reproduction)



Biological basis of behaviour

Behaviour is associated with activity in the nervous system

- E.g., measure electrical activity in nerves during behaviour, stimulating nerves causes behaviour

Behaviours can change with biological changes in the nervous system

- E.g., effects of brain tumours, lesions, drugs, etc

Behaviours often change with developmental changes to the individual

- Behaviours emerge and change predictably across a lifespan

Behaviours can run in families

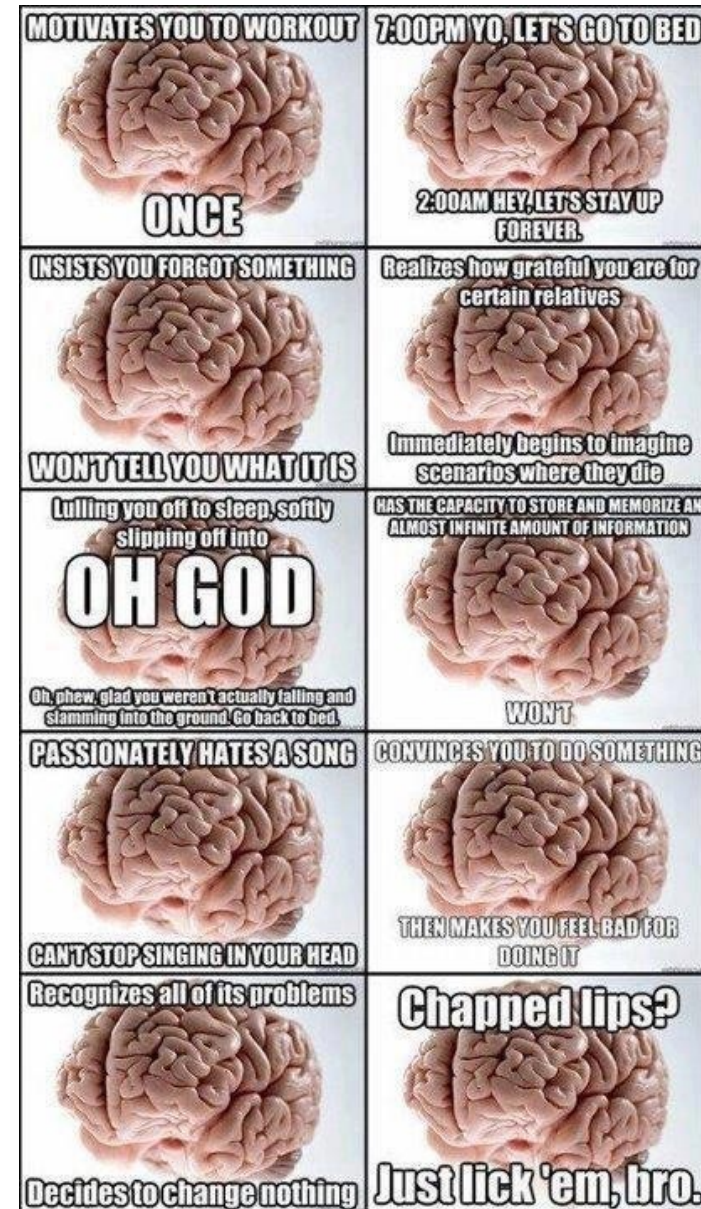
- Evidence of heritability of some behaviours

Behaviour is often species specific

- Appears in all members of the species
- Cross-cultural universality

Behaviour often has an evolutionary history:

- Homologous behaviours are often shared by related species.



Nature vs Nurture

False dichotomy between environments and genes:

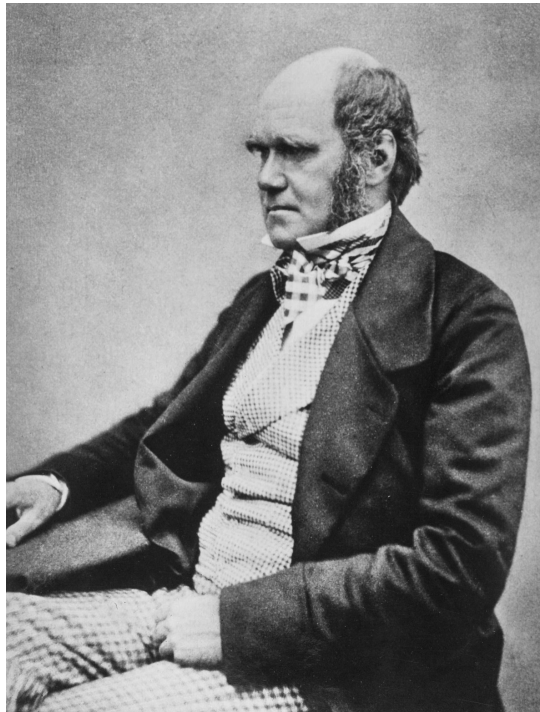
- You cannot have any behaviour without the underlying neurobiological mechanisms
- All biological mechanisms are in their present form due to the environment

A very politically loaded question

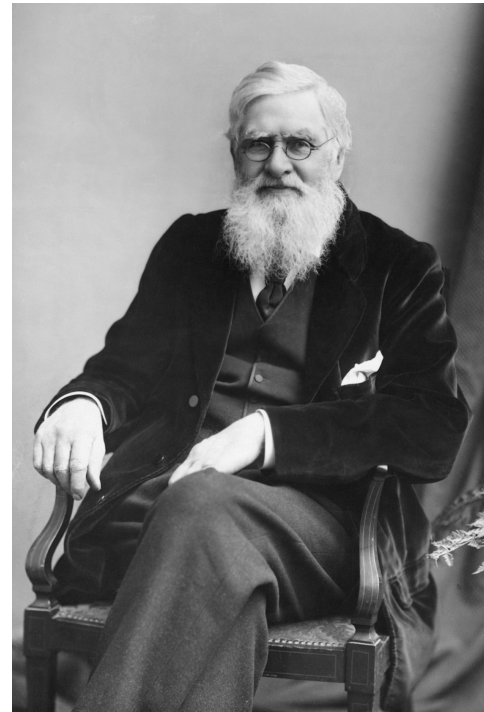
- To what degree can peoples' lives be improved by changes in social and developmental conditions?
- Just-so theories/justifications of social class:
 - E.g., Some people are poor because they are innately inferior.
- What happens if authorities acquire the ability to identify individuals with certain innate traits?

Evolution by descent and selection

Charles Darwin, 1809-1882

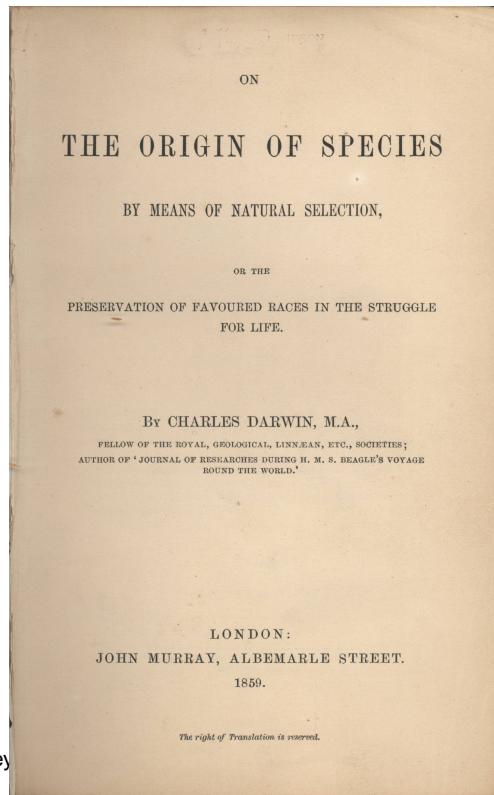


Alfred Russel Wallace, 1823-1913

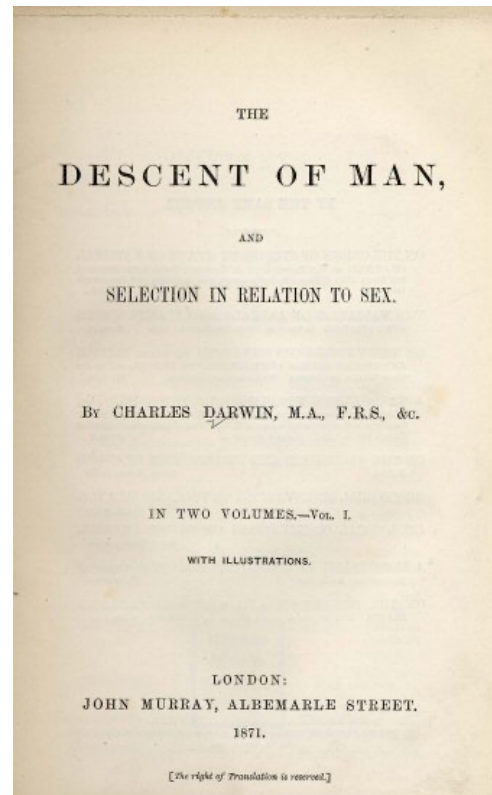


Darwin's books

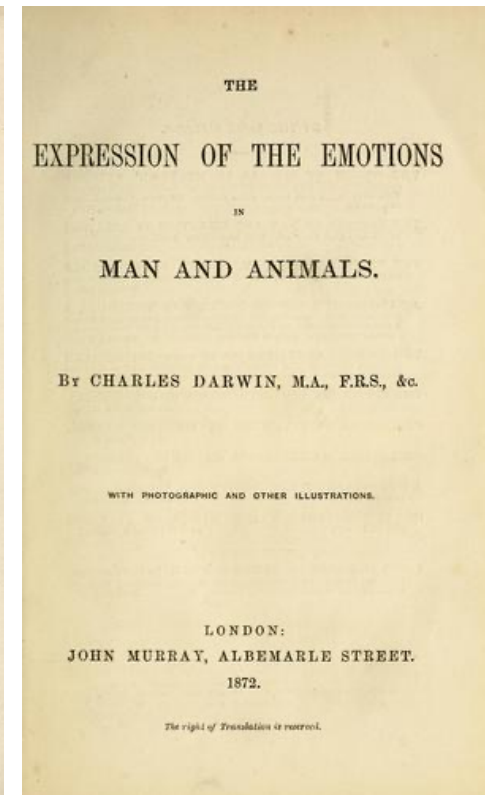
*On the Origin of Species,
1859*



*The Descent of Man, and
Selection in Relation to Sex,
1871*

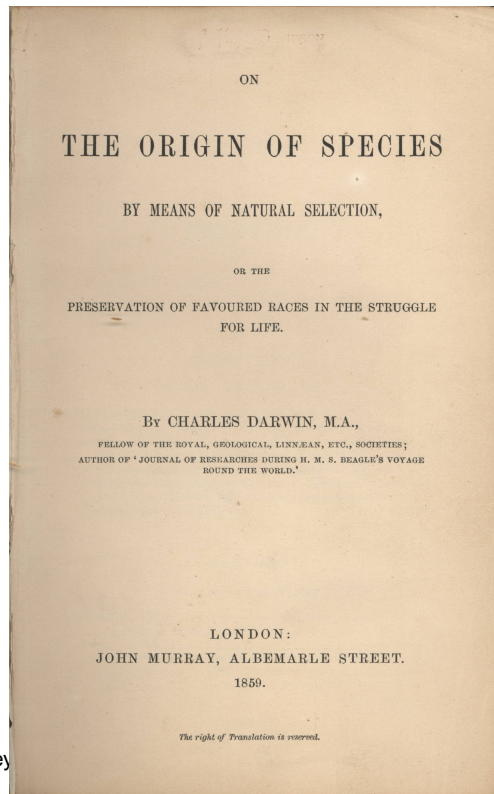


*The Expression of the
Emotions in Man and
Animals, 1872*



Darwin's books

On the Origin of Species,
1859



The origins of the Origin of the Species

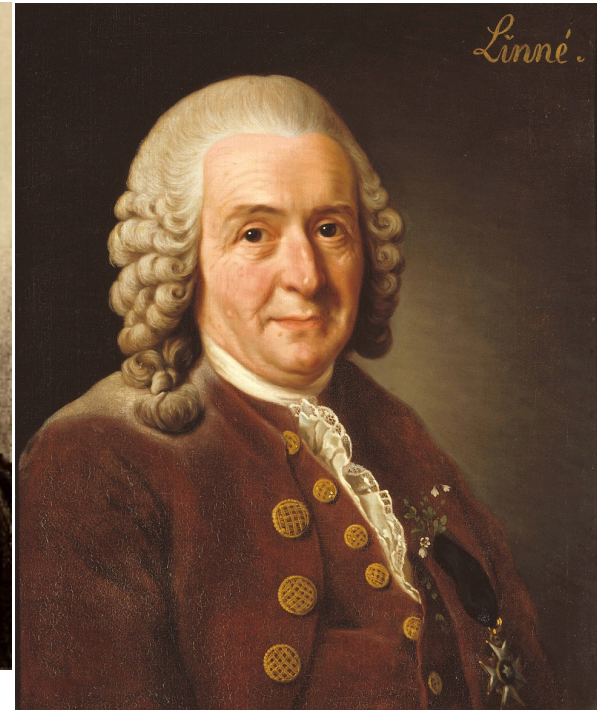
Charles Lyell, 1797-1875



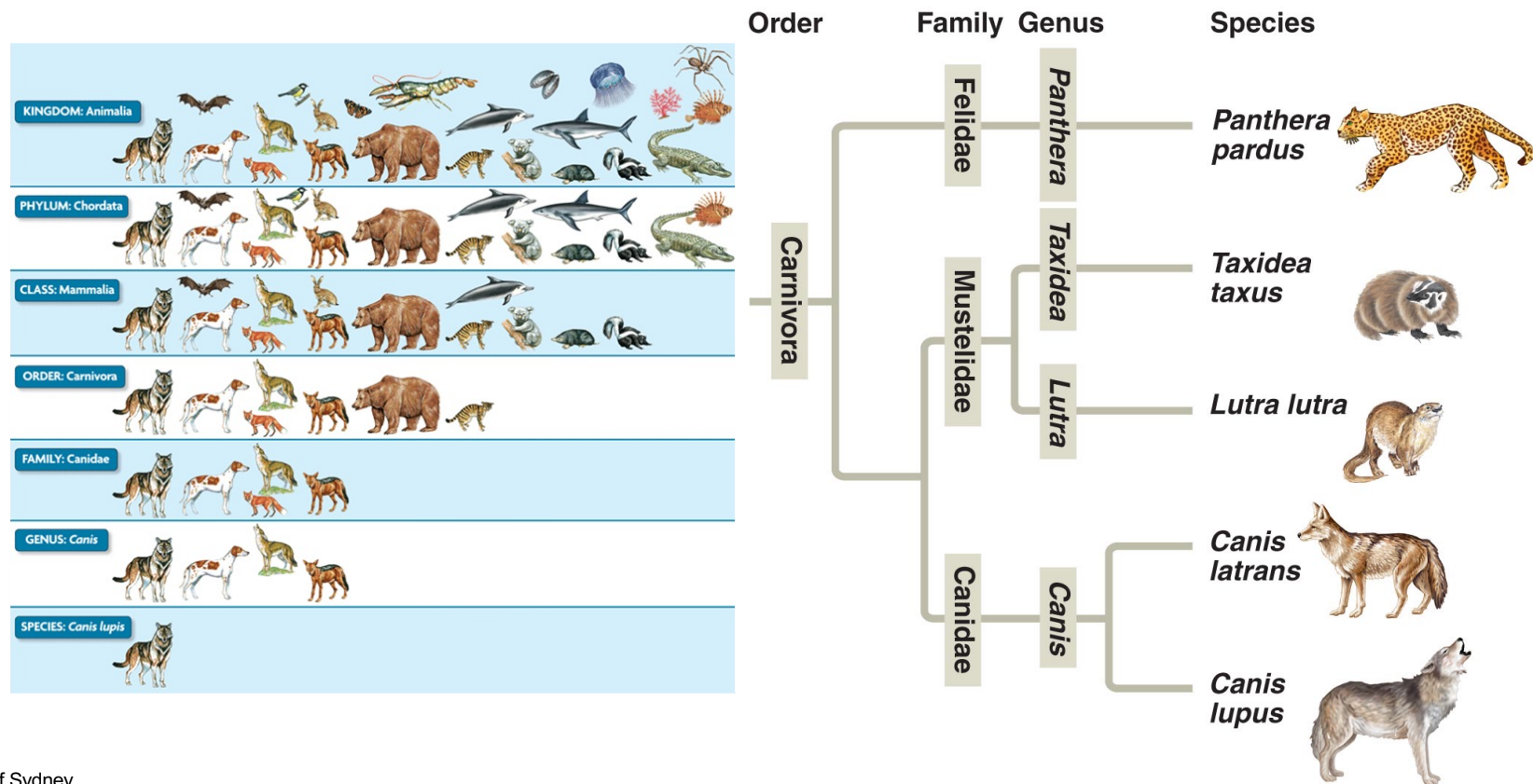
Thomas Robert Malthus, 1766-1834



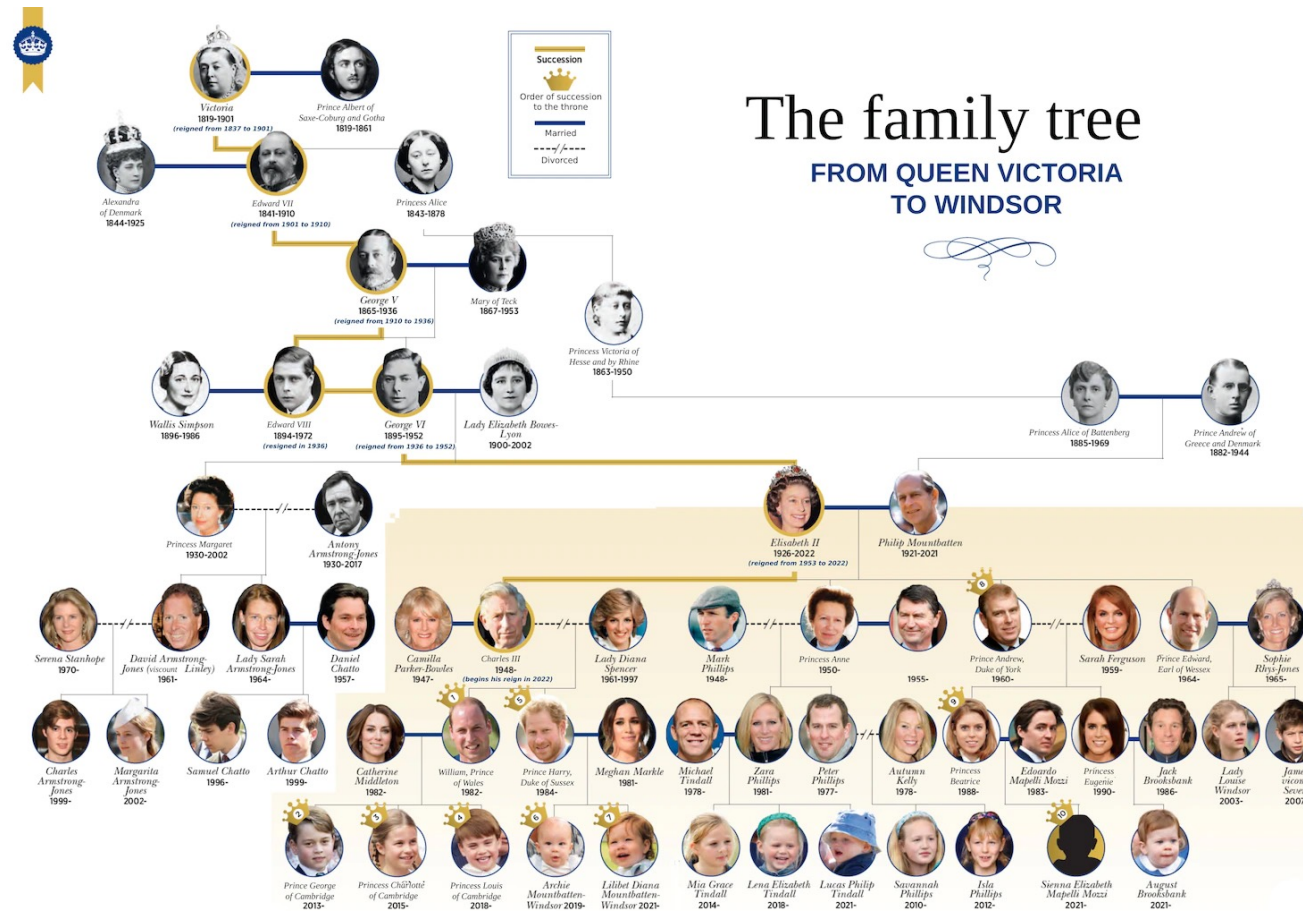
Carl Linnaeus, 1707-1778



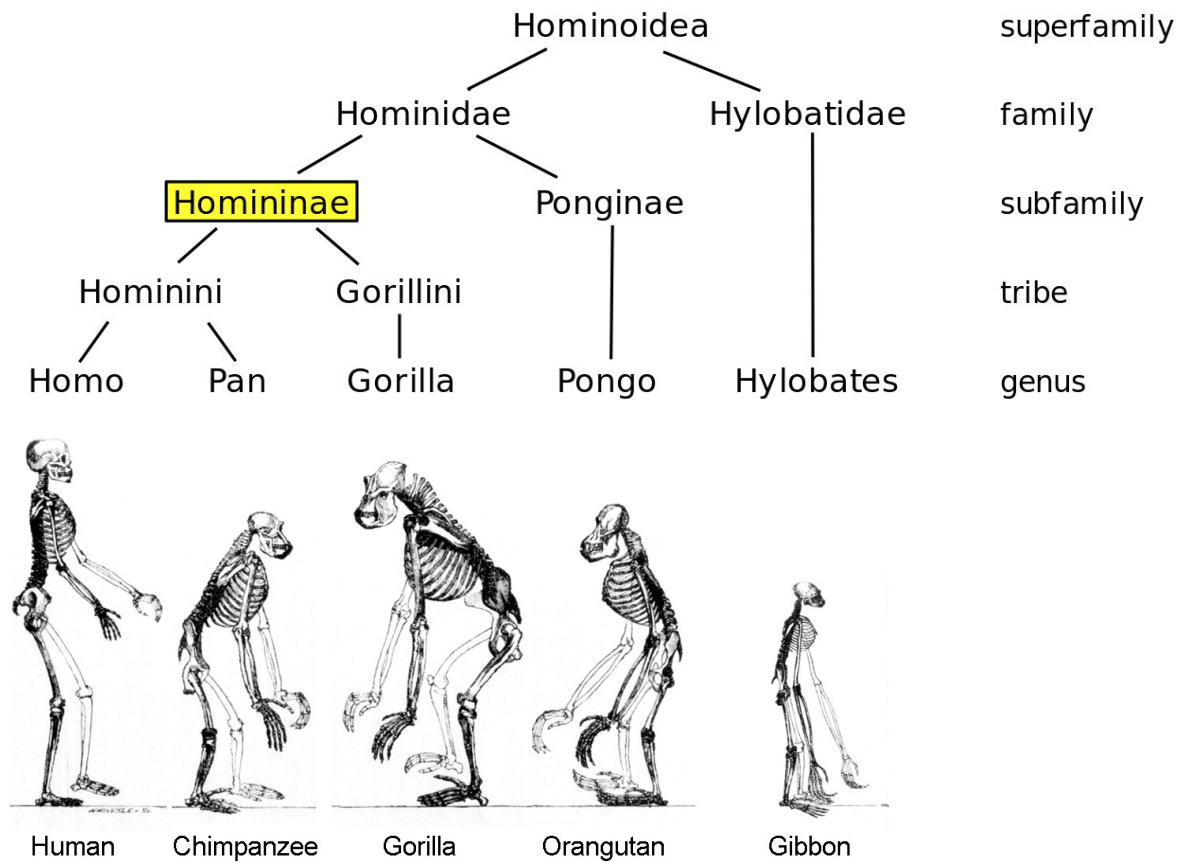
Linnaean taxonomy – sorting organisms by morphological similarities



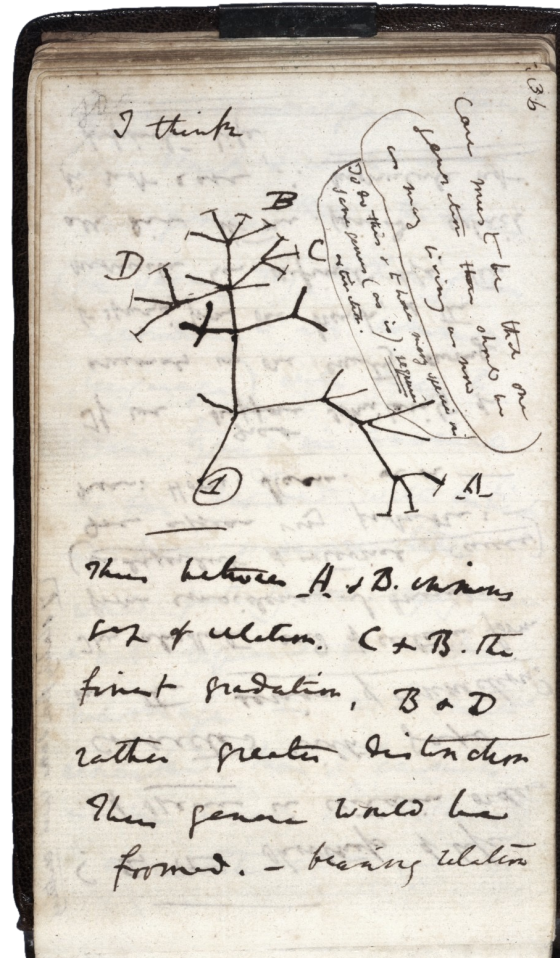
Darwin: Common morphology is because of common descent



Darwin: Common morphology is because of common descent



Common descent and continuity between species



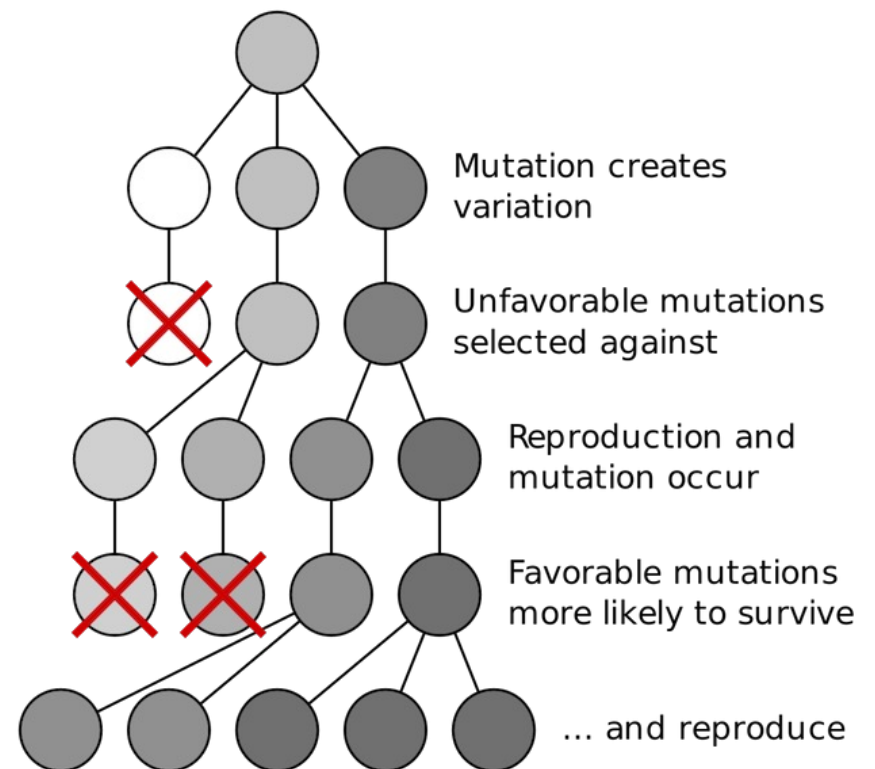
Darwin's theory of evolution by natural selection

Numerous observations that organisms evolved over time.

What was not known was how this occurred.

Darwin's theory relied on three main factors:

1. Variation of traits between individuals
2. Selection of traits that conferred reproductive success
3. Heritability of these traits



Natural selection

- **The survival of the fittest**

1. There are limited resources & there is competition for these resources,
2. Individuals who are best suited to their environment will survive and reproduce,
3. They will pass on their characteristics to their offspring (via inheritance).

- **Fitness, from an evolutionary point of view, is relative reproductive success**

- Relative to direct competitors in the population

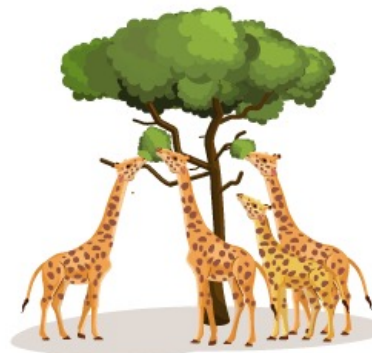
1. Variation in neck length



2. Struggle for existence



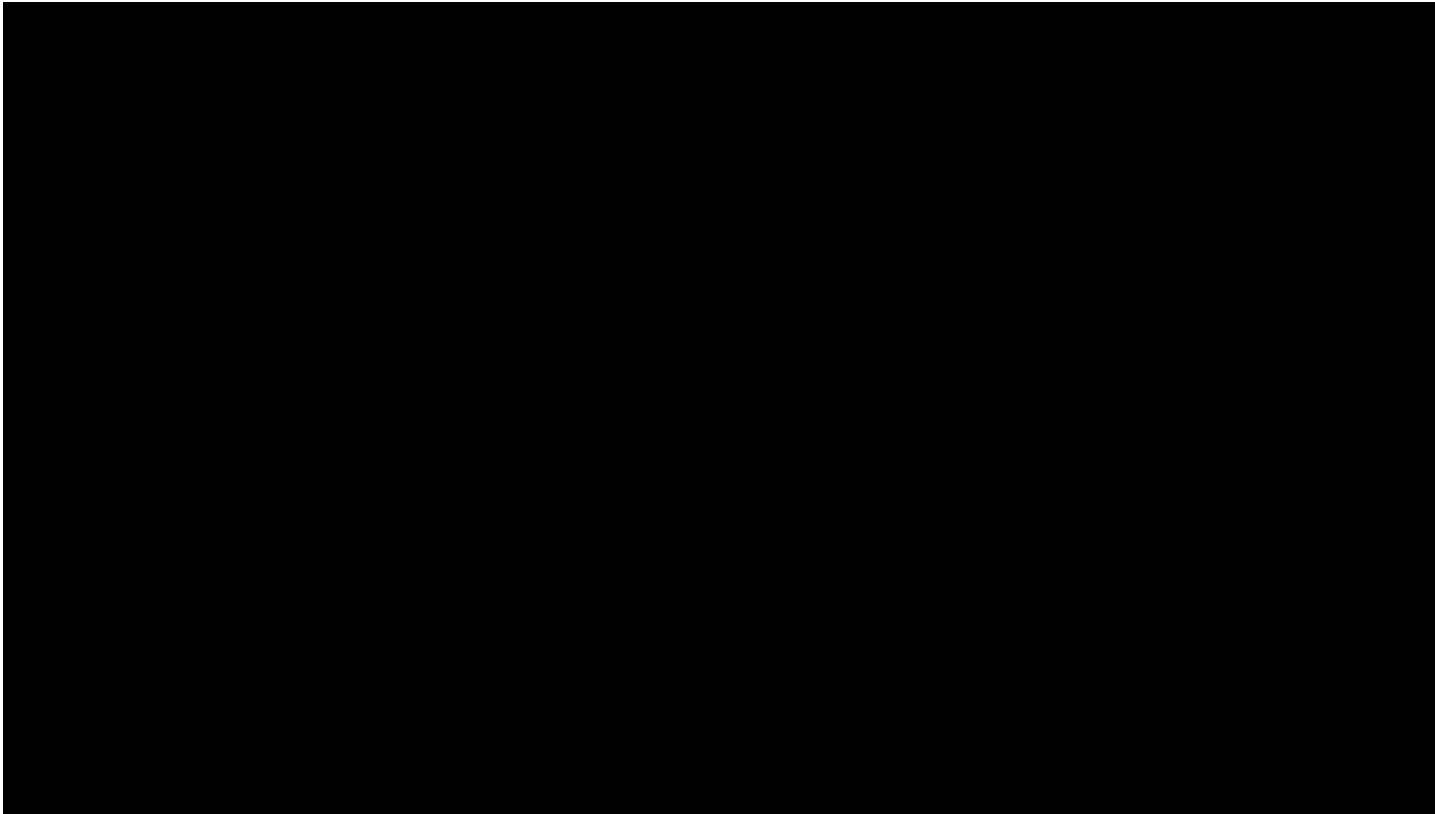
3. Greater fitness



4. Long-neck trait increases



Evolution of ciprofloxacin resistance in *E. coli*



Does natural selection occur with mental traits?

The fundamental elements of Evolutionary Psychology

Variation

- Mental traits can change subtly from individual to individual within a population

Selection

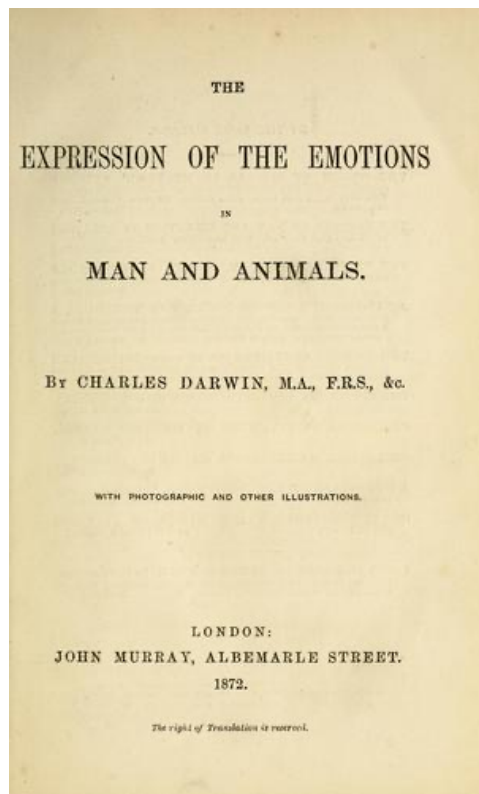
- Some variants of mental traits provide an advantage to the individual in certain environments

Inheritance

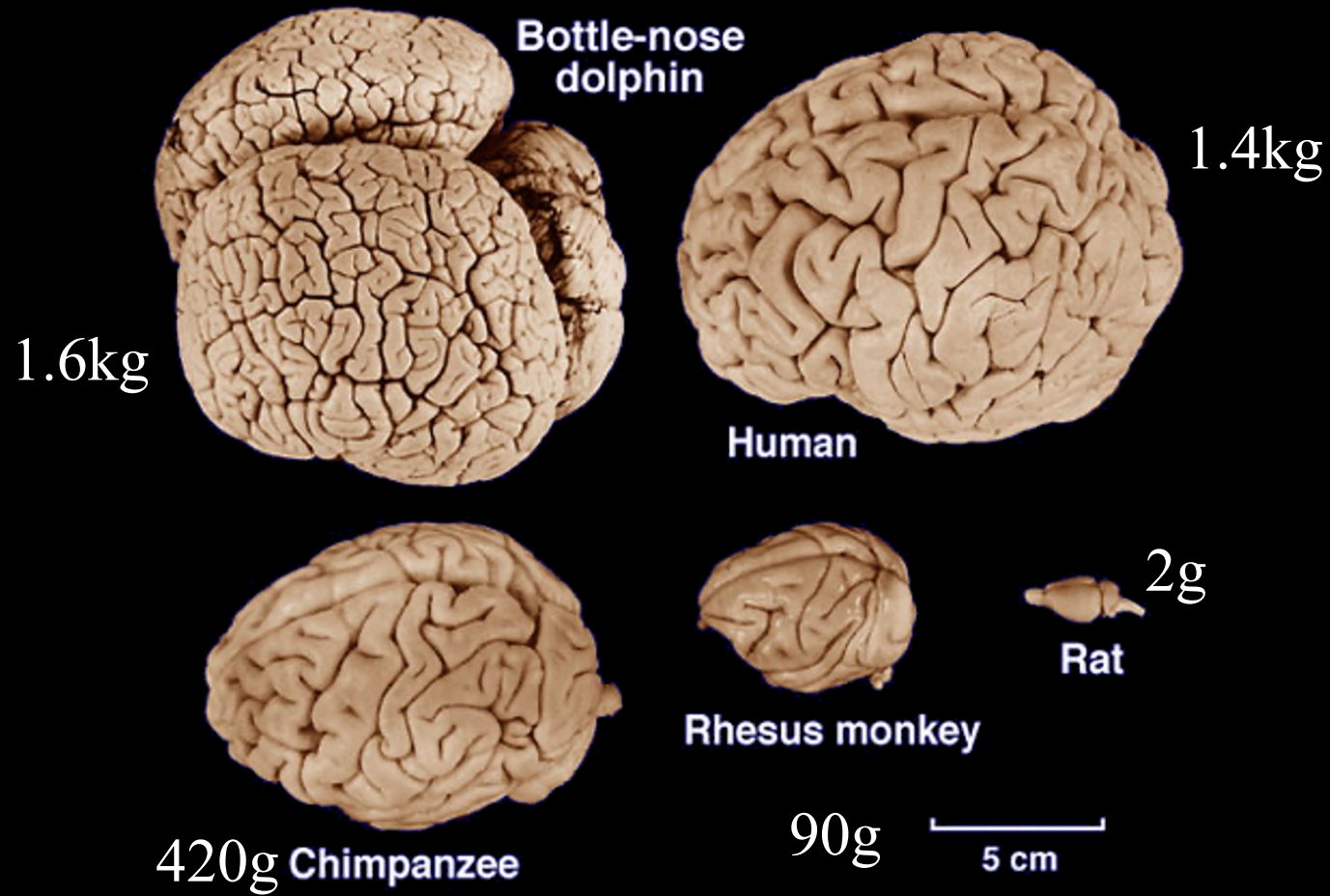
- Only variations that are inherited take place in the evolutionary process

Comparative psychology

*The Expression of the
Emotions in Man and
Animals, 1872*



Comparative neuroanatomy



Behaviours and fitness

Some behaviours have a very obvious relationship to fitness

Confer a differential reproductive advantage in their niche

Consider the species-specific defensive responses (SSDRs)

E.g., the deer mice of Washington, *Peromyscus maniculatus*.

Two sub-species

P.m. austerus – woodland regions of West Washington state

P.m. gambeli – desert regions of East Washington state

Bred in lab – no experience with predators

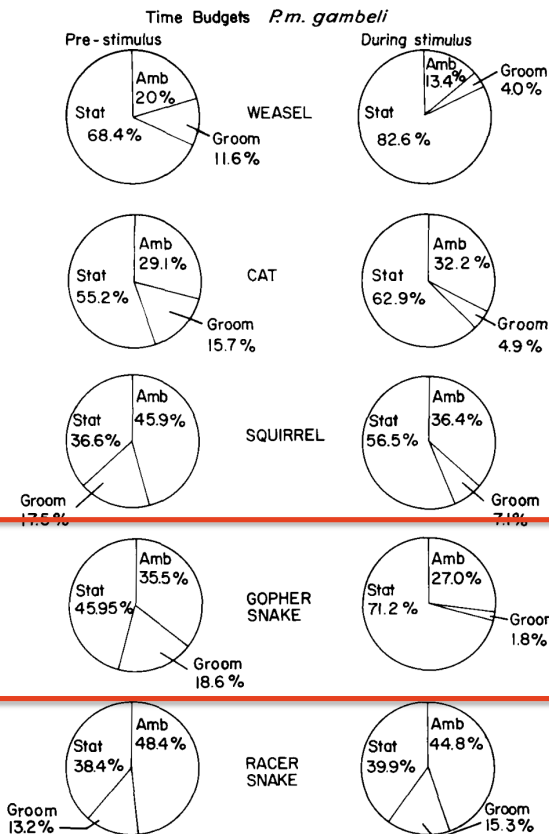


Hirsch and Bolles 1980.

<https://doi.org/10.1111/j.1439-0310.1980.tb01064.x>

Behavioural responses to potential predators

P.m. gambeli - desert



Big increase in time stationary and decrease in grooming in response to natural predators from ecological niche.

No change to non-natural predators or to non-predators (squirrel)

Fig. 1: *P. m. gambeli* time budgets for the periods before (pre-stimulus) and during the presence of stimulus animals. Each budget represents the proportion of time devoted to ambulation (Amb), grooming (Groom), and stationary posture (Stat) for each of the stimulus animal conditions

P.m. austeri - woodland

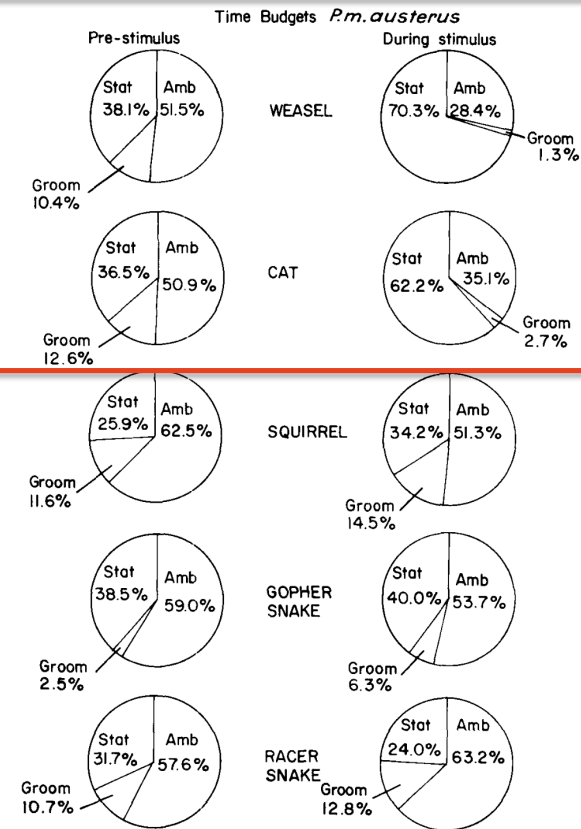


Fig. 2: *P. m. austeri* time budgets for the periods before (pre-stimulus) and during the presence of stimulus animals. Each budget represents the proportion of time devoted to ambulation (Amb), grooming (Groom), and stationary posture (Stat) for each of the stimulus animal conditions

Change in survival times when fed to predator (Gopher snake)

Table 1: Mouse vulnerability to a gopher snake $\pm$ S.D.

	Mean No. of s to first strike	Mean No. of strikes to kill	Mean No. of s to kill
<i>gambeli</i>	42.4 $\pm$ 17.48	4.3 $\pm$ 0.50	1769.9 $\pm$ 348.47
<i>austerus</i>	28.5 $\pm$ 5.13	1.9 $\pm$ 0.23	99.9 $\pm$ 33.01

P.m. austerus (desert mouse) showed defensive behaviours that were absent from *P.m. gambeli* (woodland):

1. Avoidance – they often moved out of striking range when the snake approached;
2. Escape – a vertical leap (similar to a kangaroo mouse) to escape a strike when it occurred.

P.m. gambeli often stopped to engage in grooming in the presence of the predator.

What are the benefits to fitness of an innate ability to recognize predators?

Differential survival rates means that defensive traits are more likely to be passed to next generations

Being able to recognise threats quickly means that more time can be allocated to growth behaviours

Feeding

Resting

Reproduction

Artificial selection

What Crops Looked Like Before Thousands of Years of Selective Breeding



watermelon



corn



carrot



banana

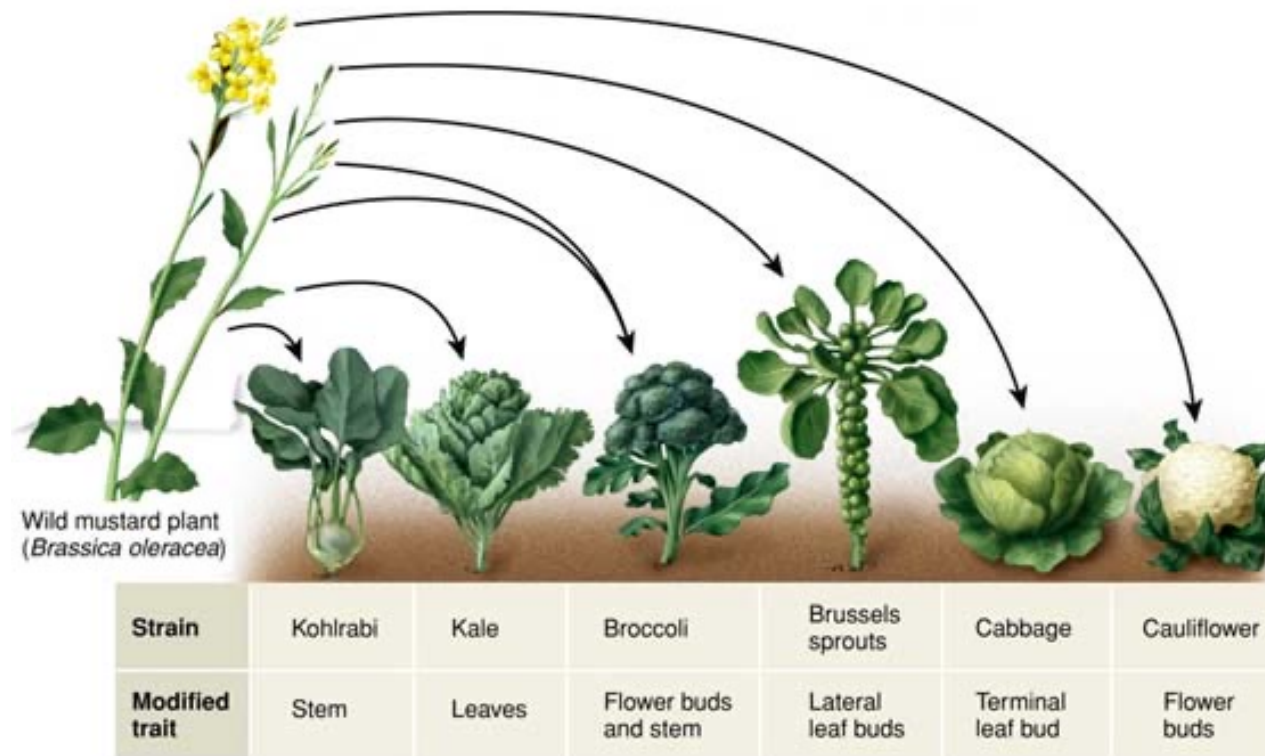


aubergine / eggplant



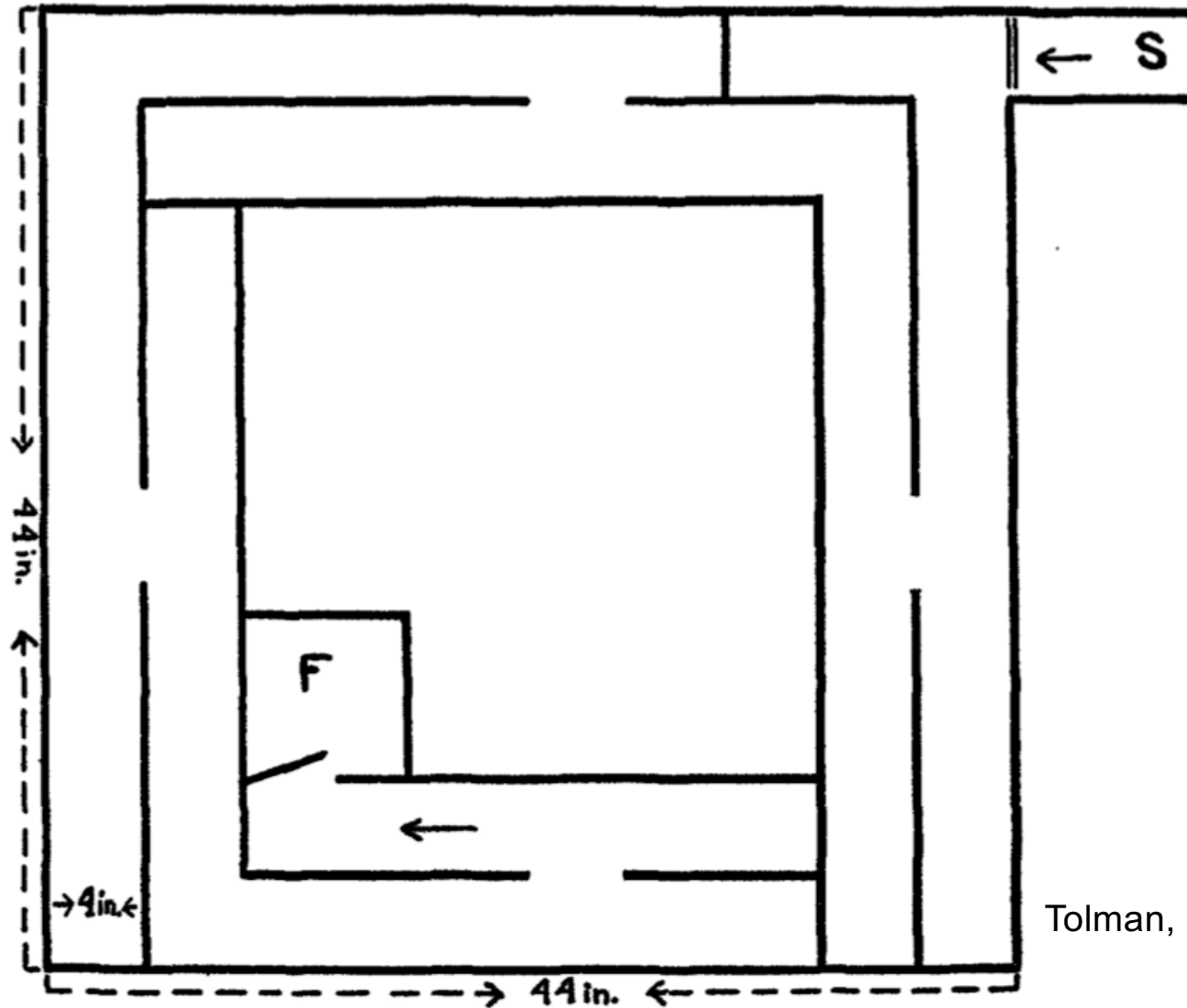
cabbage, kale, broccoli, etc.

Artificial selection



Selective breeding of cognitive phenotypes

Figure 1.



Tolman, 1924

Tryon, 1940

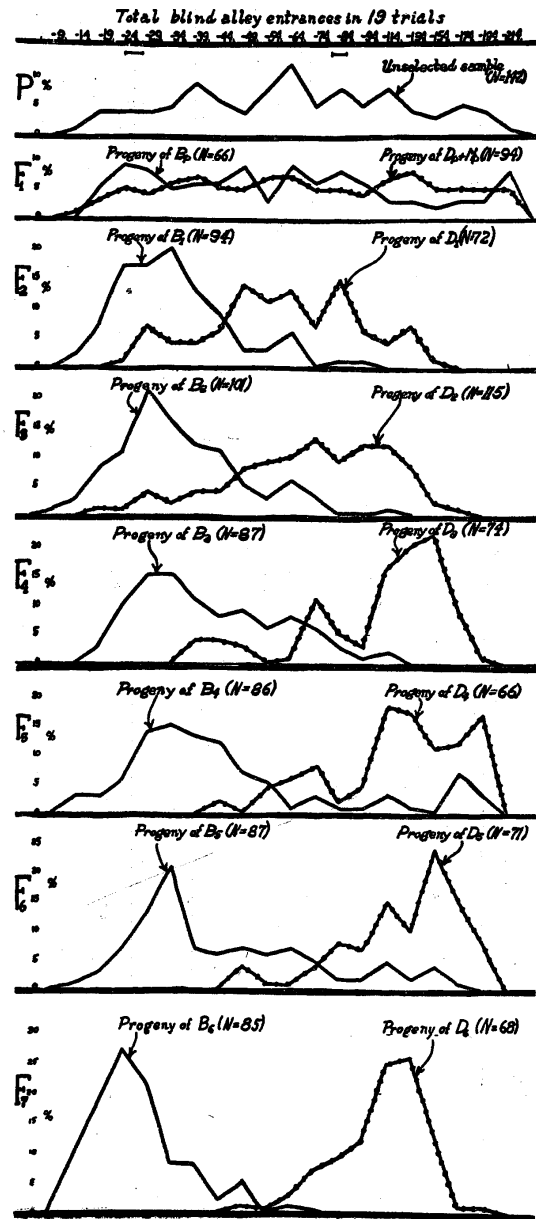


FIG. IV. — EFFECTS OF SELECTIVE BREEDING ON MAZE-LEARNING

Along the top is the scale of brightness as evidenced by the total number of blind-alley entrances made in nineteen trials. All the distributions below use this common top scale. For instance, a bright animal who made from ten to fourteen errors would fall under the scale step, the upper limit of which is marked —14, a dull who made from 195 to 214 errors would fall under —214, etc. The first generation of rats, marked "P" to the left, is shown just below the scale. The total number of P rats was 142, and the percent of them lying at each point on the scale is indicated in the distribution. The brightest of these were bred together, and then the dullest, giving the two F_1 groups, as shown. The selective breeding effects are shown down to the F_7 , where progeny of B_6 (bright F_6) are markedly different from the progeny of D_6 (dull F_6).

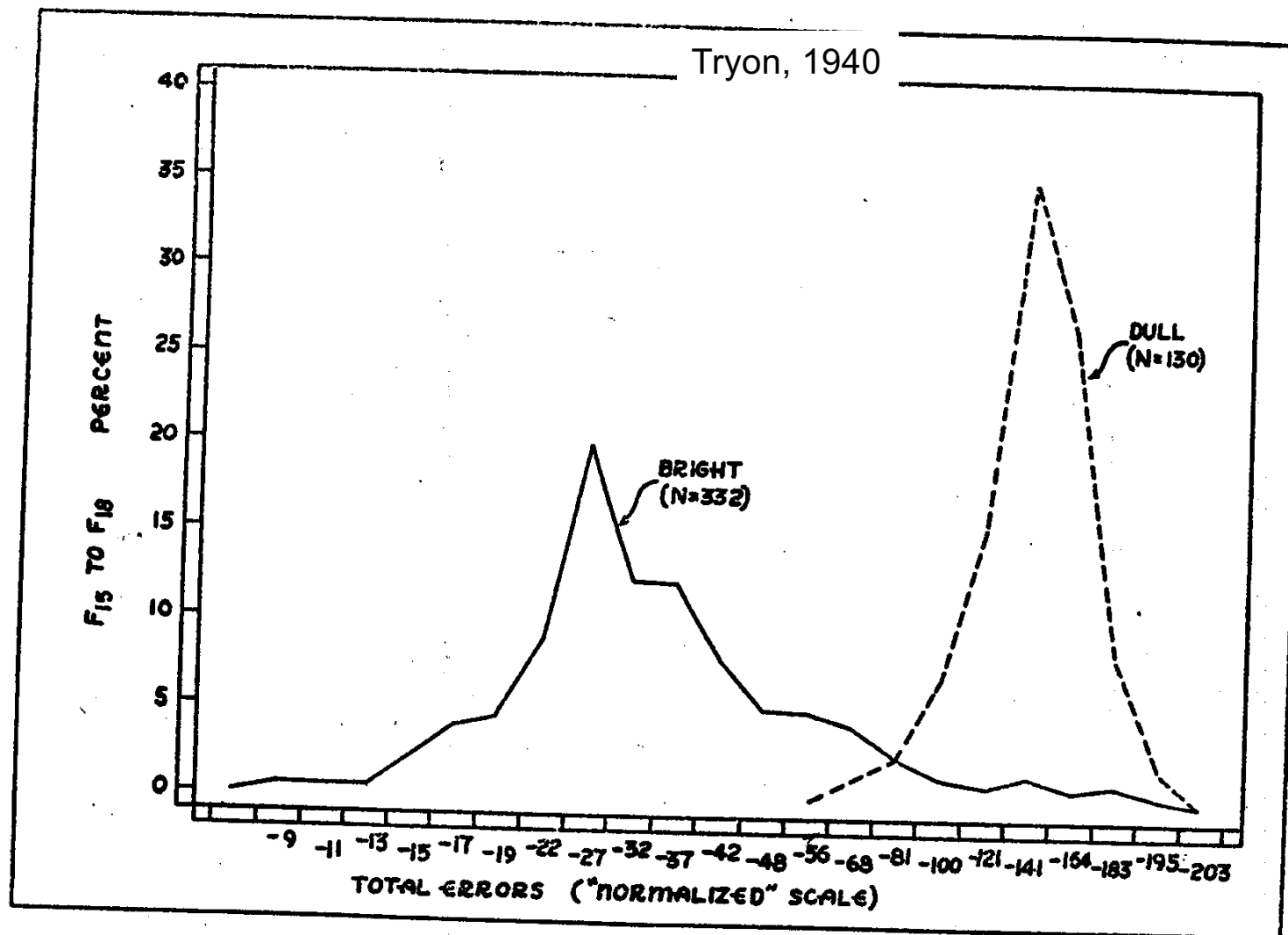


FIG. V. — BRIGHT AND DULL STRAINS OF THE FIFTEENTH TO EIGHTEENTH GENERATIONS

Misuse of Evolutionary Psychology

It has been used to justify:

- Aggression:

 - Within communities/groups

 - Between communities/groups

 - Within sexes

 - Between sexes

- Social Darwinism

 - Applying evolutionary ideas to economics and society - Individuals and groups compete to succeed in a 'survival of the fittest'

- Structural inequality:

 - Inter-generational wealth occurs not because of unequal access to opportunities but because certain families contain sets of traits that gives them advantages

- Eugenics

 - Artificial selection to change traits at the population level

Behaviours and fitness

It is easy to see how some mental traits
increase fitness:

Aggression and competition for resources

Intelligence and acquisition of resources
and/or avoiding danger

But what of others?

E.g., the following universal human traits:

Love

Cooperation

Parental care

Music and dancing



Inclusive fitness

- **From an evolutionary point of view, success of a trait is determined by:**
 - the number of copies of that trait that are propagated into successive generations.
- **Typically seen as an example of personal or individual fitness:**
 - The number of offspring that an individual begets
- **However, all that matters is that the number of individuals carrying the trait in subsequent generations increases**
 - It does not matter if they are your own offspring.
- **Total fitness is:**
 - Direct fitness (own offspring) +
 - Indirect fitness (others' offspring who hold the same trait)
- **Hamilton (1964) argued that altruistic traits could survive if altruistic individuals preferentially increased the fitness of any offspring who held the same trait.**
- **This is inclusive fitness:**
 - The number of offspring carrying the same trait that an individual supports into the next generation (regardless of who begets them).

What might inclusive fitness help explain?

– Mate selection

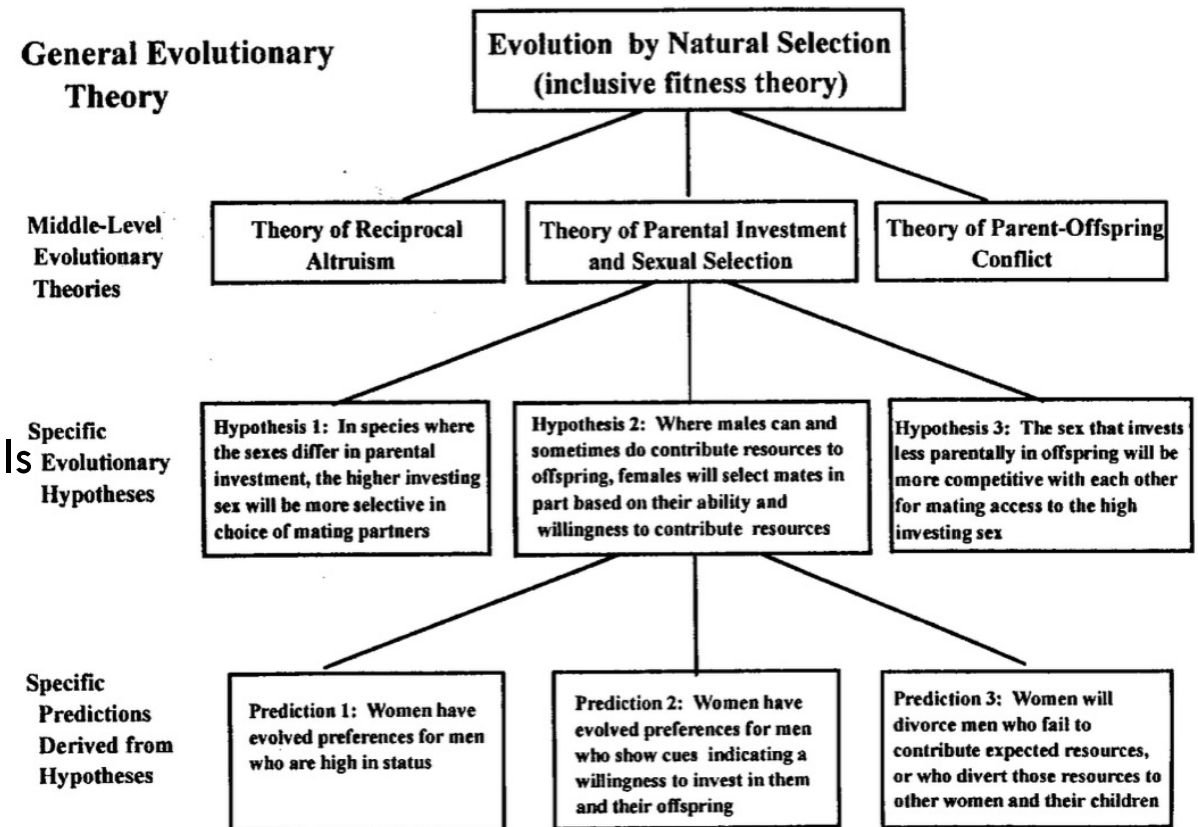
- Markers of health
- Markers of parental investment

– Kin favouritism/selection

- Parental investment
- Altruism towards related individuals

– Cooperation and altruism

- Cohesive social groups
- Cognitive traits to detect cheating



Readings

Natural selection of mental traits:

- Scholarpedia, http://www.scholarpedia.org/article/Evolutionary_psychology
- Cosmides and Tooby, Evolutionary Psychology, a primer, <https://www.cep.ucsb.edu/wp-content/uploads/2023/06/Evolutionary-Psychology-A-Primer-CosmidesTooby1993.pdf>
- Buss, D (2009) American Psychologist, 64: 140-148.
http://pzacad.pitzer.edu/~dmoore/2009_Buss_AmPsycho_Darwin.pdf